

Mid-Semester Test Solution

Q1. $[4 \times 2 = 8 \text{ Marks}]$.

- (a) $m(t) = \text{sinc}(\pi t)$; energy in frequency domain $= \int_{-1/2}^{1/2} \Pi^2(f) = 1$.
- (b) Envelop of signal $\sqrt{m^2(t) + m_h^2(t)}$.
- (c) Using $10^6 = \frac{3L^2}{1000}$, number of levels in PCM $L = 1.875 \times 10^4$, and $n = \log_2(L) = 14.1562 = 15$ bits per sample. For DPCM: $10^6 = \frac{3L^2}{10}$, number of levels in PCM $L = 1.8257 \times 10^3$, and $n = \log_2(L) = 10.8343 = 11$ bits per sample. Savings of 4 bits per sample [**1 Mark**]. Saving in cost: $(2 \times 20,000 \times 4)/10 = 16000 \text{ Rs}$ [**1 Mark**].
- (d) BW of upper signal $= 100 \times 3 = 300 \text{ Hz}$. BW of lower signal $= 250 + 200 = 450 \text{ Hz}$. After the summer, the maximum frequency to the sampler input is 450 Hz [**1 Mark**]. Nyquist interval: $1/900 \text{ seconds}$ [**1 Mark**].

Q2. Using cosine sum formula, express the modulated signal as $x(t) = 40 \cos 1000\pi t + 20 \cos 1000\pi t \cos 200\pi t = (40 + 20 \cos 200\pi t) \cos 1000\pi t$. $[4 \times 2 = 8 \text{ Marks}]$.

- (a) Since $x(t) = (40 + 20 \cos 200\pi t) \cos 1000\pi t$, message signal $m(t) = \cos 200\pi t$, carrier signal $= 40 \cos 1000\pi t$, and sideband signal $= 20 \cos 200\pi t \cos 1000\pi t$. Carrier power $P_c = 40^2/2 = 800$, sideband power $P_s = 100$. Power efficiency $= 100/900 = 0.1111 = 11.11\%$.
- (b) Modulation index $\mu = 20/40 = 1/2 = 50\%$
- (c) Message signal $m(t) = \cos 200\pi t$, Message BW $= 100 \text{ Hz}$. Modulated BW $= 200 \text{ Hz}$.
- (d) Message signal $m(t) = \cos 200\pi t$, carrier signal $c(t) = 40 \cos 1000\pi t$; PM signal $= 40(\cos 1000\pi t + k_p m(t)) = 40(\cos 1000\pi t + 50\pi \cos 200\pi t)$.

Q3. $m(t) = 10 \cos(100\pi t)$, $f_m = 50$; carrier $\cos(2000\pi t)$ or $\sin(2000\pi t)$, $k_f = 25$. FM signal $\cos(2000\pi t + 2\pi k_f \int_0^t m(\tau) d\tau) = \cos(2000\pi t + 5 \sin 100\pi t)$ or FM signal $\sin(2000\pi t + 2\pi k_f \int_0^t m(\tau) d\tau) = \sin(2000\pi t + 5 \sin 100\pi t)$. $[4 \times 2 = 8 \text{ Marks}]$.

- (a) $f_i(t) = \frac{1}{2\pi} \frac{d(\theta)}{dt} = 1000 + 250 \cos(100\pi t)$. $\Delta_f = 250$. $\beta = \Delta_f/f_m = 250/50 = 5$. BW $= 2(\Delta_f + f_m) = 2(250 + 50) = 600 = 2f_m(\beta + 1) = 100 \times 6 = 600 \text{ Hz}$.
- (b) Input FM signal $\cos(2000\pi t + 5 \sin 100\pi t)$, output to the channel $= \frac{1}{5} \cos(2000\pi t + 5 \sin 100\pi t)$; Input FM signal $\sin(2000\pi t + 5 \sin 100\pi t)$, output to the channel $= \frac{1}{5} \sin(2000\pi t + 5 \sin 100\pi t)$.
- (c) The filter is a differentiator: Output at the filter: $\frac{d}{dt} \frac{1}{5} \cos(2000\pi t + 5 \sin 100\pi t) = -\frac{1}{5} \sin(2000\pi t + 5 \sin 100\pi t) \times (2000\pi + 500\pi \cos(100\pi t))$ or $\frac{d}{dt} \frac{1}{5} \sin(2000\pi t + 5 \sin 100\pi t) = \frac{1}{5} \cos(2000\pi t + 5 \sin 100\pi t) \times (2000\pi + 500\pi \cos(100\pi t))$.
- (d) Envelop detector output: $= +\frac{1}{5}(2000\pi + 500\pi \cos(100\pi t))$. Output at the DC blocker: $+100\pi \cos(100\pi t)$

Q4. $[4 \times 2 = 8 \text{ Marks}]$.

- (a) $h(t) = u(t) - u(t - 1)$ or $h(t) = \Pi(t - \frac{1}{2})$, where $u(t)$: unit step function, and $\Pi(\cdot)$ is rectangular pulse.
- (b) $H(f) = \text{sinc}(\pi f) e^{-j\pi f}$
- (c) The sketch is given in Figure 6:

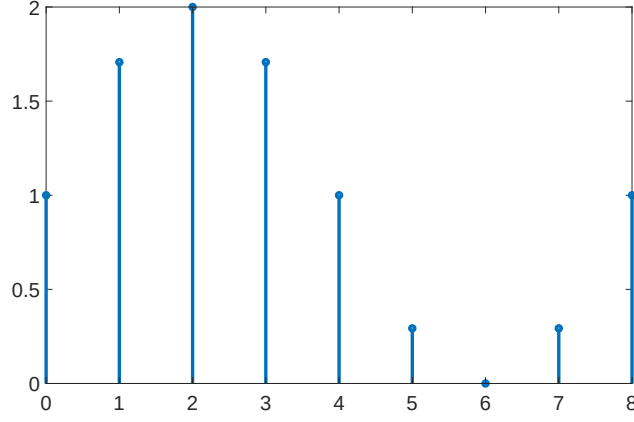


Figure 6: Solution: Q4(c)

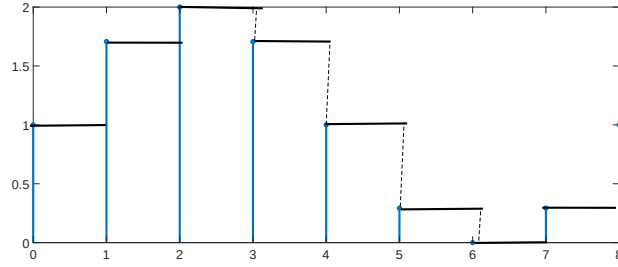


Figure 7: Solution: Q4(d)

(d) The sketch will be similar to Figure 6 with a rectangular pulse corresponding to sample values. It is shown in Figure 7: 4(d).

- Q5. (a) $T_b = \frac{1}{R_b} = \frac{1}{1000}$ [1 Mark], $p(t) = 2\Pi(\frac{t}{T_b})$, $P(f) = 2T_b \text{sinc}(\pi f T_b)$ [1 Mark]. $S_x(f) = \frac{1}{T_b}[R_0 + 2 \sum_{n=1}^{\infty} R_n \cos 2\pi f T_b]$. where $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$ is the auto-correlation of the line encoded waveform. $R_0 = 4$, $R_n = 0, n > 0$. $S_x(f) = \frac{1}{T_b} R_0 = \frac{4}{T_b}$ [1 Mark]. $S_y(f) = P(f)^2 S_x(f) = \frac{16T_b^2 \text{sinc}^2(\pi f T_b)}{T_b} = 16T_b \text{sinc}^2(\pi f T_b) = \frac{16}{1000} \text{sinc}^2(\frac{\pi f}{1000})$ [1 Mark].
- (b) Data file size 2GB = 16×10^9 bits. Bit rate 8Mbps. Symbol rate = $2 \times 200 = 400$ Ksymbols/sec [2 Marks]. Number of bits per symbol = $(8 \times 10^6)/(400 \times 10^3) = 20$ bits/symbol [1 Mark]. Number of symbols = $(16 \times 10^9)/20 = 8 \times 10^8$ [1 Mark].
- Q6. (a) $m_p = 5 = A, L = 2^5 = 32$, SQNR = $\frac{3L^2 P_m}{m_p^2}$ [1 Mark]; In the quarter period of signal $m(t) = 4At/T_0$, where $T_0 = 0.01$, $P_m = \frac{4}{T_0} \int_0^{T_0/4} m(t)^2 dt = A^2/3 = 25/3$. [2 Marks]. SQNR = $\frac{3L^2 P_m}{m_p^2} = (3 \times 32^2 \times 25)/(25 \times 3) = 1024 = 30.1030\text{dB}$ [1 Mark].
- (b) SQNR = $\frac{3L^2}{[\log(1+\mu)]^2} = \frac{3 \times 32^2}{[\log(1+100)]^2}$ [1 Mark] = 144.2298 = 21.5906 dB [1 Mark]. The logarithm is base e i.e. natural logarithm.
- (c) To avoid slope overload condition $\Delta f_s \geq \frac{dm(t)}{dt}$. Using step size minimum sampling frequency $0.1f_s = \frac{dm(t)}{dt}$ [1 Mark]. Since $m(t) = 4At/T_0$. The slope of signal $\frac{dm(t)}{dt} = 4A/T_0 = 20/0.01 = 2000$ [2 Marks]; Using $0.1f_s = 2000$, $f_s = 20,000$ Hz [1 Mark].